Problem 1 – Differential sensitivity Regularized Learning Task (50 Points)

Motivated by the role of sensitivity in privacy, memorization, and generalization, in this problem, we study differential sensitivity regularization problems. In particular, we consider

$$\min_w f(w) + \rho\, \mathbb{E}_z \left\|\nabla_z \ell(w,z)\right\| \tag{1}$$

and

$$\min_w f(w) + \rho\, \mathbb{E}_z \left\|\nabla_w \ell(w,z)\right\| \tag{2}$$

where $f(w) = \mathbb{E}[\ell(w,z)]$ is the average loss and $w \in \mathbb{R}^d$ and $z \in \mathbb{R}^p$.

The first problem could help us with mitigating memorization and improving robustness while the second problem could help us in improving generalization by reducing the sharpness of loss landscape.

For each problem, we want to calculate efficient approximation of gradients to be used in training algorithms such as SGD

a) For the first problem, note that

$$\nabla_w \|\nabla_z \ell(w,z)\| = \nabla^2_{wz} \ell(w,z) \frac{\nabla_z \ell(w,z)}{\|\nabla_z \ell(w,z)\|}. \tag{3}$$

Let $u \sim \mathcal{N}(0, I_d) \in \mathbb{R}^d$. Using zero-order approximation (i.e., utilizing finite differences with random perturbation) of each columns of $\nabla^2_{wz}\ell(w,z)$ with smoothing parameter $\mu > 0$ show that

$$\nabla^2_{wz} \ell(w,z) \approx \frac{u}{\mu} \Big(\nabla_z \ell(w+\mu u, z) - \nabla_z \ell(w,z) \Big)^\top \tag{4}$$

and thus

$$\nabla_w \|\nabla_z \ell(w,z)\| \approx \frac{\langle \nabla_z \ell(w+\mu u, z), \nabla_z \ell(w,z)\rangle - \|\nabla_z \ell(w,z)\|^2}{\mu \|\nabla_z \ell(w,z)\|} u \tag{5}$$

b) For the second problem, note that

$$\nabla_w \|\nabla_w \ell(w,z)\| = \nabla^2_{w} \ell(w,z) \frac{\nabla_w \ell(w,z)}{\|\nabla_w \ell(w,z)\|}. \tag{6}$$

Let $u = \frac{\nabla_w \ell(w,z)}{\|\nabla_w \ell(w,z)\|}$ and note that by first-order Taylor approximation

$$\nabla^2_w \ell(w,z) u \approx \nabla_w \ell(w+u,z) - \nabla_w \ell(w,z). \tag{7}$$

using this result argue that

$$\nabla_w \ell(w+\rho u, z), \qquad u = \frac{\nabla_w \ell(w,z)}{\|\nabla_w \ell(w,z)\|} \tag{8}$$

is a good estimator of the gradient of the objective of the second problem, i.e. $f(w) + \rho\, \mathbb{E}_z \left\|\nabla_w \ell(w,z)\right\|$.

c) Assume $\ell(w, z)$ is L-smooth w.r.t. w for any z, i.e.,

$$\|\nabla \ell(w_1, z) - \nabla \ell(w_2, z)\|_2 \leq L\|w_1 - w_2\|_2, \qquad \forall z.$$

Consider the update: $w_{t+1} = w_t - \eta g_t$ for $t = 1, \ldots, T$ where the update vector is given by

$$g_t = \nabla \ell \left(w_t + \rho_t \frac{\nabla \ell(w_t, z_t)}{\|\nabla \ell(w_t, z_t)\|_2}, z_t \right),$$

with $z \overset{\text{i.i.d.}}{\sim} p_z$ for some learning rates $\eta \geq 0$ and $\rho_t \geq 0$. Note that while η is kept fixed across the iterations, ρ_t varies.

Assume the stochastic gradient $\nabla \ell(w_t, z_t)$ satisfies the following assumptions,

$$\mathbb{E}_{z_t} \left[\nabla \ell(w_t, z_t) \right] = \nabla \mathcal{L} \left(w_t \right), \quad \mathbb{E}_{z_t} \left[\|\nabla \ell(w_t, z_t) - \nabla \mathcal{L} \left(w_t \right)\|_2^2 \right] \leq \sigma^2,$$

for all $t = 1, \ldots, T$ where $\sigma > 0$ and $\mathbb{E}_{z_t}$ denotes the expectation w.r.t. z_t conditioned on $z_1, \ldots, z_{t-1}$.

Show that under the above assumptions it holds that

$$\mathbb{E}_{z_t} \|g_t\|_2^2 \leq 2L^2 \rho_t^2 + 2\|\nabla f(w_t)\|_2^2 + 2\sigma^2.$$

Problem 2 – Convergence of Switching Gradient Descent (50 Points)

In this problem, we are interested in analyzing SGM for a simple setting. Notably, consider the constrained learning task

$$w^* = \arg\min_w f(w) \qquad s.t. \quad h(w) \leq 0 \tag{9}$$

Recall SGM performs $w_{t+1} = w_t - \eta g_t$ where $g_t = \nabla f(w_t)$ if $h(w_t) \leq \epsilon$ and $g_t = \nabla h(w_t)$ if $h(w_t) > \epsilon$. We assume both f and h are convex and G-Lipschitz.

a) Using the analysis GD for convex and G-Lipschitz functions show that

$$\frac{1}{T}\sum_{t=1}^{T}\langle g_t, w_t - w^*\rangle \leq \frac{\eta G^2}{2} + \frac{D^2}{2\eta T} \tag{10}$$

where $D = \|w_1 - w^*\|$.

b) Let

$$[T] = \{1, \ldots, T\}, \qquad A = \{t \in [T] | h(w_t) \leq \epsilon\}, \qquad B = [T] \backslash A \tag{11}$$

Using the convexity of f and h show that

$$\frac{1}{T}\sum_{t\in A}(f(w_t) - f(w^*)) + \frac{1}{T}\sum_{t\in B} h(w_t) \leq \frac{\eta G^2}{2} + \frac{D^2}{2\eta T} \tag{12}$$

c) Define the output of SGM by $\bar{w} = \frac{1}{|A|}\sum_{t\in A} w_t$ where $|A|$ is the cardinality of set A. Using convexity of h show that $h(\bar{w}) \leq \epsilon$.

d) Let $\eta = \frac{D}{G\sqrt{T}}$ and $\epsilon = \frac{DG}{\sqrt{T}}$. Show that $f(\bar{w}) - f(w^*) \leq \epsilon$.

These steps show that we can find a near optimal solution which is approximately feasible as well.